

TUNG CHANG, Thailand

WMO: 483070

Lat: 19.4082N

Lon: 100.8826E

Elev: 335

StdP: 97.37

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.1	12.8	9.4	7.6	13.9	11.0	8.5	15.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	9.7	36.8	24.9	35.6	24.9	34.5	24.9	27.2	32.2	26.8	31.7	26.6	31.3	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	22.3	29.5	25.7	21.8	29.0	25.4	21.5	28.6	88.4	32.2	86.9	31.9	85.5	31.5	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	8.0	38.2	1.8	1.7	6.6	39.4	5.5	40.4	4.5	41.3	3.2	42.6		
(5)			8.1	28.1	2.1	0.6	6.6	28.5	5.4	28.9	4.2	29.2	2.7	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.8	21.3	23.2	25.7	28.1	28.5	28.1	27.2	27.0	27.3	26.5	24.6	21.8
	DBStd	2.92	2.23	1.82	1.98	1.75	1.64	1.38	1.49	1.19	1.05	1.24	1.85	2.18
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	9	6	0	1	0	0	0	0	0	0	0	0	3
	CDD10.0	5759	349	371	487	542	572	542	533	526	519	513	438	367
	CDD18.3	2726	97	137	229	292	314	292	275	268	269	254	188	111
	CDH23.3	25213	749	1306	2374	3169	3274	2993	2529	2242	2319	2033	1416	808
CDH26.7	10191	215	573	1251	1691	1534	1133	810	627	779	785	549	243	
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	1587	21	12	39	109	194	200	307	360	249	79	30	15
	PrecMax	1940	80	64	103	179	375	443	434	732	428	204	127	103
	PrecMin	1282	0	0	0	26	84	69	203	247	143	27	0	0
	PrecStd	218	27	18	29	39	69	89	67	117	85	42	35	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.3	34.6	36.7	39.1	38.2	35.7	35.5	33.3	33.7	33.8	33.4	31.8
		MCWB	22.2	23.1	23.7	24.1	25.4	26.3	25.9	26.7	26.5	25.5	25.1	23.2
	2%	DB	30.2	33.3	35.6	37.4	36.3	34.1	33.1	31.9	32.8	33.0	32.2	30.6
		MCWB	21.6	23.0	23.6	24.7	25.8	26.2	26.1	26.3	26.3	25.5	24.4	22.4
	5%	DB	29.1	32.1	34.5	35.9	34.7	32.9	31.8	30.9	31.9	32.1	31.3	29.4
		MCWB	21.2	22.4	23.5	25.0	25.9	26.1	26.0	26.1	26.1	25.5	24.0	22.1
	10%	DB	27.8	30.6	33.2	34.4	33.3	31.7	30.6	29.9	30.8	30.9	30.1	28.0
		MCWB	20.8	21.8	23.1	24.9	25.8	26.0	25.9	25.9	25.9	25.2	23.8	21.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.0	25.8	27.4	27.5	27.2	27.2	27.2	27.4	27.4	27.1	26.6	24.9
		MCDB	28.7	32.4	35.1	34.5	32.9	32.1	31.7	31.3	31.9	32.0	31.5	30.3
	2%	WB	23.0	24.3	25.6	26.5	26.8	26.8	26.7	26.8	26.9	26.4	25.5	23.9
		MCDB	27.7	30.8	32.8	33.5	32.7	31.7	31.0	30.4	31.4	31.1	30.4	28.6
	5%	WB	22.2	23.4	24.5	25.9	26.4	26.5	26.4	26.5	26.5	26.0	24.9	23.0
		MCDB	26.9	30.0	31.7	32.6	32.3	31.1	30.3	29.9	30.6	30.4	29.7	27.1
	10%	WB	21.4	22.5	23.7	25.3	26.0	26.2	26.0	26.1	26.1	25.5	24.3	22.1
		MCDB	26.0	28.8	31.0	31.8	31.8	30.5	29.7	29.3	29.9	29.6	28.6	26.4
Mean Daily Temperature Range	5% DB	MDBR	13.8	16.1	15.7	13.0	9.7	7.3	6.3	6.2	7.6	9.5	11.8	13.0
		MCDBR	15.6	18.1	17.9	15.4	12.2	9.2	8.6	7.8	9.2	11.1	13.0	14.4
		MCWBR	7.3	7.8	6.8	4.6	3.4	2.6	2.7	3.0	3.3	4.4	5.8	6.8
	5% WB	MCDBR	13.2	16.3	16.2	13.4	10.6	8.2	7.4	7.2	8.3	9.8	11.5	12.5
		MCWBR	6.7	8.1	7.3	4.8	3.4	2.8	2.7	3.0	3.4	4.3	5.5	6.8
Clear-Sky Solar Irradiance	taub	0.451	0.563	0.762	0.723	0.514	0.457	0.454	0.478	0.503	0.542	0.437	0.434	
	taud	2.147	1.850	1.473	1.593	2.135	2.327	2.327	2.242	2.151	1.998	2.257	2.222	
	Ebn at Noon	820	744	620	654	798	837	841	826	804	753	829	823	
	Edh at Noon	143	203	307	274	158	129	129	141	153	173	127	129	
All-Sky Solar Radiation	RadAvg	4.38	5.14	5.29	5.75	5.24	4.66	4.27	4.25	4.54	4.51	4.52	4.13	
	RadStd	0.27	0.29	0.30	0.39	0.48	0.35	0.38	0.27	0.19	0.20	0.32	0.27	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (9 neighbors)	+0.30	N/A	N/A	N/A	N/A	+0.19	N/A	+1	+134	+137	

Nomenclature: See separate page